Quantitative Easing and the U.S. Housing Market: The Federal Reserve's Transmission Channel in the Post-COVID Economy (2020–2022)

Summary

In this paper, I will examine the post-pandemic surge in home prices and the role monetary policy played in driving it. Between 2020 and 2022, home prices rose by 34% (Figure 1), driven primarily by three factors: expansive fiscal policy, pandemic-induced shifts in housing preferences, and monetary policy. While my focus will be on monetary policy, the first two factors also had significant effects. First, the unprecedented scale of fiscal stimulus sharply increased household savings and strengthened balance sheets, enabling more households to afford home purchases. Second, the shift to remote work led to large-scale migration from dense urban centers to suburban areas, further boosting housing demand. The core of this paper will focus on quantitative easing (QE) and be spread out into 3 main parts: 1) the macro-fundamentals of QE and housing, 2) an analytical synopsis of 2020 – 2022 displaying the transmission mechanism discussed in part 1 and 3) if policy response (QE) was appropriate in hindsight. I will explain how Federal Reserve's large-scale purchases of agency mortgage-backed securities (MBS) and treagsuries helped push mortgage rates to record lows, contributing to one of the sharpest increases in home prices in modern U.S. history. Additionally, I will argue that while the immediate implementation of QE was appropriate to calm markets, the continuation of the large QE program created distortions in the housing market.

Macro-Fundamentals: Quantitative Easing -> Mortgage Rates -> Housing Demand

From a macroeconomic perspective, the IS–LM framework helps illustrate the logic behind QE (Figure 2). When the Fed purchases long-term bonds, it raises their prices and lowers their yields, thereby reducing long-term borrowing costs. As rates fall, households and firms can obtain cheaper financing, leading to greater investment and higher spending. The result is an increase in overall aggregate demand and output, particularly in interest-sensitive sectors such as housing and durable goods. Ben Bernanke put the idea of QE elegantly: "Federal Reserve purchases of mortgage-backed securities (MBS), for example, should raise the prices and lower the yields of those securities...declining yields and rising asset prices ease overall financial conditions and stimulate economic activity through channels similar to those for conventional monetary policy (Bernanke, 2012)." In practice, the mortgage rate can be viewed as the yield on newly issued MBS plus a spread that reflects servicing costs and the premium required to originate mortgages (Fuster et al.). As the Fed's purchases drove down MBS yields, the reduction typically passed through almost one-for-one to lower mortgage rates. Moreover, the Federal Reserve's purchases were not limited to MBS but also included U.S. Treasuries. MBS generally trades at a spread over Treasury yields, the simultaneous decline in both Treasury and MBS yields pushed mortgage rates even lower. Mortgage rates are the largest driver of housing demand. Small changes in mortgage rates alter monthly payments disproportionately. An example of this can be seen in Figure 3, which shows how lowering mortgage rates from 5 percent to 3 percent on the median single-family house (~$410,000) with a 20 percent down payment reduces the monthly payment by roughly 21 percent. Anenberg and Ringo (Anenberg et al) show that housing supply responds only gradually to demand shocks, so short-run housing volatility is driven mainly by demand. The slow response of supply is intuitive, given that permitting, construction, and final sales are lengthy processes. Consequently, monetary policy affects housing primarily through the mortgage-rate channel (housing demand), which rapidly alters buyer entry and price growth (Anenberg et al.).

Post-COVID (2020-2022) Analysis

The scope of the Federal Reserve's quantitative easing program in the post-COVID period was substantial. Between March 2020 and March 2022, the Fed's ~$4 tn balance sheet more than doubled, reaching $9 tn (Figure 4). At its peak, the Federal Reserve owned roughly 32 percent of the aggregate market (Klein et al.). During this period, agency MBS issuance increased by approximately $1.5 trillion (Figure 5), of which the Fed purchased $1.33 trillion or roughly 90 percent (Goodman et al.). In essence, the majority of agency loans originated during this time were financed directly by the Fed. As shown in Figure 6, once QE began, MBS spreads tightened by roughly 150 basis points peak-to-trough. Additionally, refinances skyrocketed as borrowers seized a once-in-a-lifetime opportunity to lock in historically low mortgage rates (Figure 7). Both 10-year Treasury yields and 30-year fixed mortgage rates fell to record lows (Figure 8). It is also important to note that this period coincided with a significant "flight to quality," as investors sought the safety of U.S. Treasuries which also lowered yields (FRB).

There are several ways to measure these dynamics: (1) by examining median home prices (Figures 9), (2) by comparing home price appreciation across different mortgage rate regimes (Figure 10), and (3) by observing the number of homes sold during the period (Figure 11, Figure 12). These indicators collectively illustrate the strength of the relationship between QE and housing market activity. As mortgage rates fell during the QE period, home prices climbed well above their pre-QE trend and sales reached close to record 2004-2005 levels (Figure 11). Zillow data shows that by April 2022, the median home value was roughly $50,000 above its pre-pandemic trend (Figure 9), reflecting the impact of cheaper borrowing costs and ample liquidity. At the same time, monthly home sales peaked around 650,000 units (Figure 12) during the height of QE before dropping sharply as mortgage rates rose above 6 percent. Home sales fell 40% from May 2022 to November 2022 as quantitative tightening (QT) was announced. The scatterplot between mortgage rates and changes in home values reinforces this relationship: lower rates are consistently associated with stronger price appreciation (Figure 10). We can see that the highest quarterly home price appreciation was when mortgage rates were below 3% from April 2020 to March 2022 while the Fed was engaging in QE. Taken together, these figures demonstrate that the 2020–2022 housing boom was largely correlated to demand-side forces via the sharp decline in mortgage rates induced by the Fed's large-scale asset purchase programs.

Ex-post, did the Federal Reserve act appropriately considering?

In hindsight, the Federal Reserve's actions were appropriate in the early stages of the pandemic but became increasingly miscalibrated as economic conditions improved. In 2020, the Fed's aggressive response (cutting rates to zero, restarting quantitative easing, and launching emergency lending facilities) was essential to stabilize financial markets and prevent a collapse in liquidity. At the onset of the crisis, market liquidity was weakening: Treasury yields were spiking, and MBS spreads were widening sharply. In April 2020, the Fed purchased $1.2 tn of securities through QE (Congressional Research Service). At that stage, policy was clearly appropriate and prevented a far deeper contraction. By 2021, however, the rationale for maintaining large-scale asset purchases had weakened considerably. Data from January 2021 through April 2022 show that the Fed was still purchasing roughly $80 billion in Treasuries and $40 billion in MBS per month. Over this period, its balance sheet expanded from $6.8 trillion to $8.5 trillion (Figure 13). In mid-2021, the economy had largely reopened, the labor market was recovering (Figure 15,16) and inflation concerns were pressing (Figure 14). Yet by continuing QE at full speed, particularly through sustained MBS purchases, the Fed kept mortgage rates artificially low and further fueled housing demand. In retrospect, the Fed should have tapered sooner and ended MBS purchases before Treasury purchases. As Woodford warned, while QE can successfully stimulate demand, it also carries financial-stability risks if maintained for too long: “There is a potential trade-off: the same policy actions that succeed in easing credit conditions may, if maintained too long, create distortions in financial markets and increase the risk of a destabilizing unwinding” (Woodford). This trade-off is precisely what unfolded in the U.S. housing market during 2021 and early 2022. My ex-post analysis of the Fed’s response can be overly critical. At the time, policymakers did not know the full extent of the virus or how long the disruptions would last. New variants were emerging quickly, and there was significant uncertainty about the depth and duration of the shock. It is easy in hindsight to say that policy was miscalibrated, but in real time the Fed was operating in an unprecedented environment.

Summary

The evidence presented in this paper supports the thesis that the Federal Reserve’s large-scale purchases of Treasuries and agency mortgage-backed securities between 2020 and 2022 significantly lowered long-term interest rates and mortgage rates, fueling one of the fastest housing booms in modern U.S. history. QE was initially an appropriate and necessary response to restore liquidity to Treasury and mortgage markets. However, as economic conditions improved through 2021, the continuation of aggressive asset purchases (mainly MBS) proved misaligned with macroeconomic fundamentals. A faster tapering of asset purchases, particularly an earlier stop to MBS purchases, could have reduced the scale of the housing surge without undermining the overall recovery. Ultimately, while QE achieved its immediate goal of stabilizing financial markets and supporting economic activity in 2020, its extended duration into 2021 and early 2022 magnified the imbalances that now define the post-pandemic housing market, specifically affordability (Figure 18). The Fed’s experience underscores the balance between supporting recovery and fueling excess demand in markets.

Sources

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<u>Figures</u>

Figure 1: Zillow Home Price Data March 2020 – March 2022 (start of QE – end of QE)

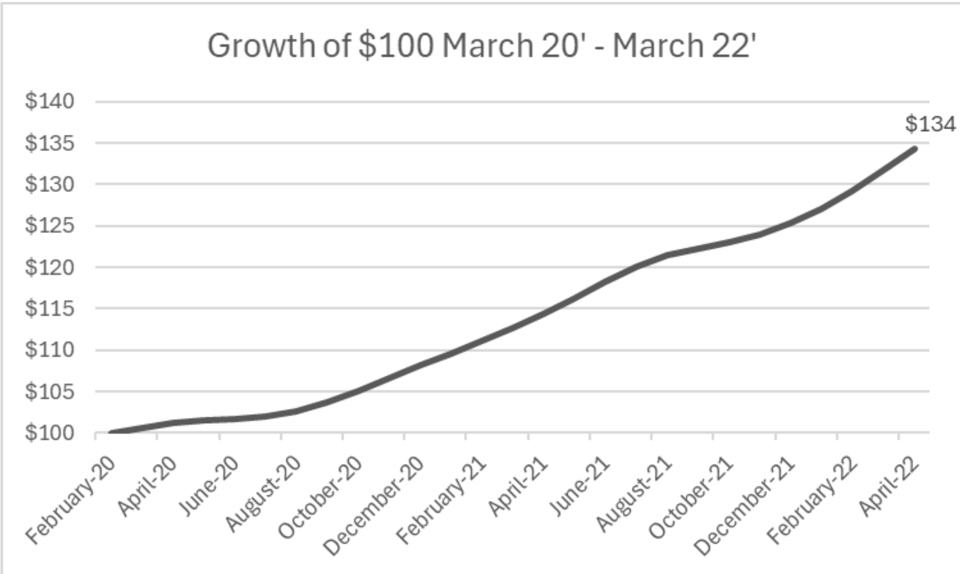

Figure 2: Illustrative Example of IS-LM Curve Diagram Under QE

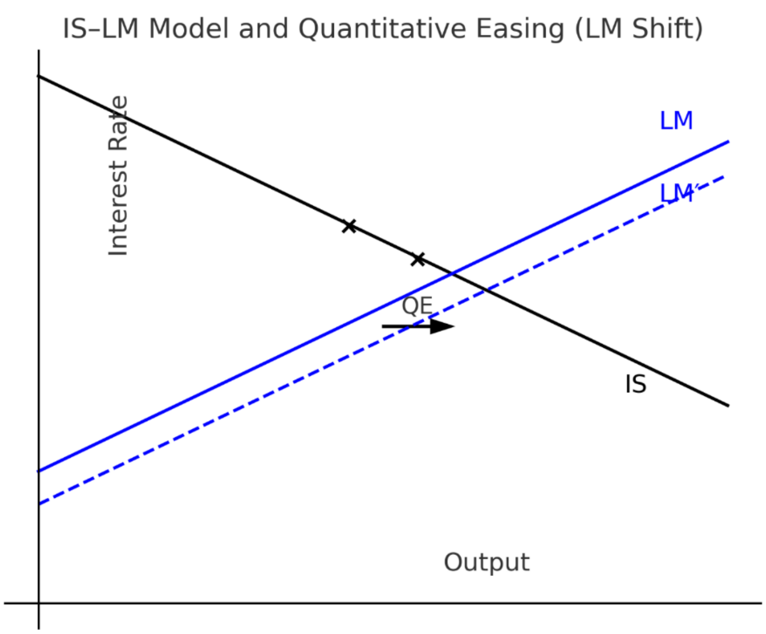

Figure 3: Illustrative example of monthly payment on median home when mortgage rates go from 5% to 3%:

	House Price ($)	Down Payment (%)	Loan Amount ($)	APR (%)	Monthly Payment ($)	Term (years)
5% Rate	$410,000	20	$328,000	5	$1,761	30
3% Rate	$410,000	20	$328,000	3	$1,383	30

Figure 4 (Federal Reserve Balance Sheet 2019 – 2025, Data via Bloomberg). Starts at roughly ~4 bn in March 2020 and peaks at ~9 bn in March 2022.

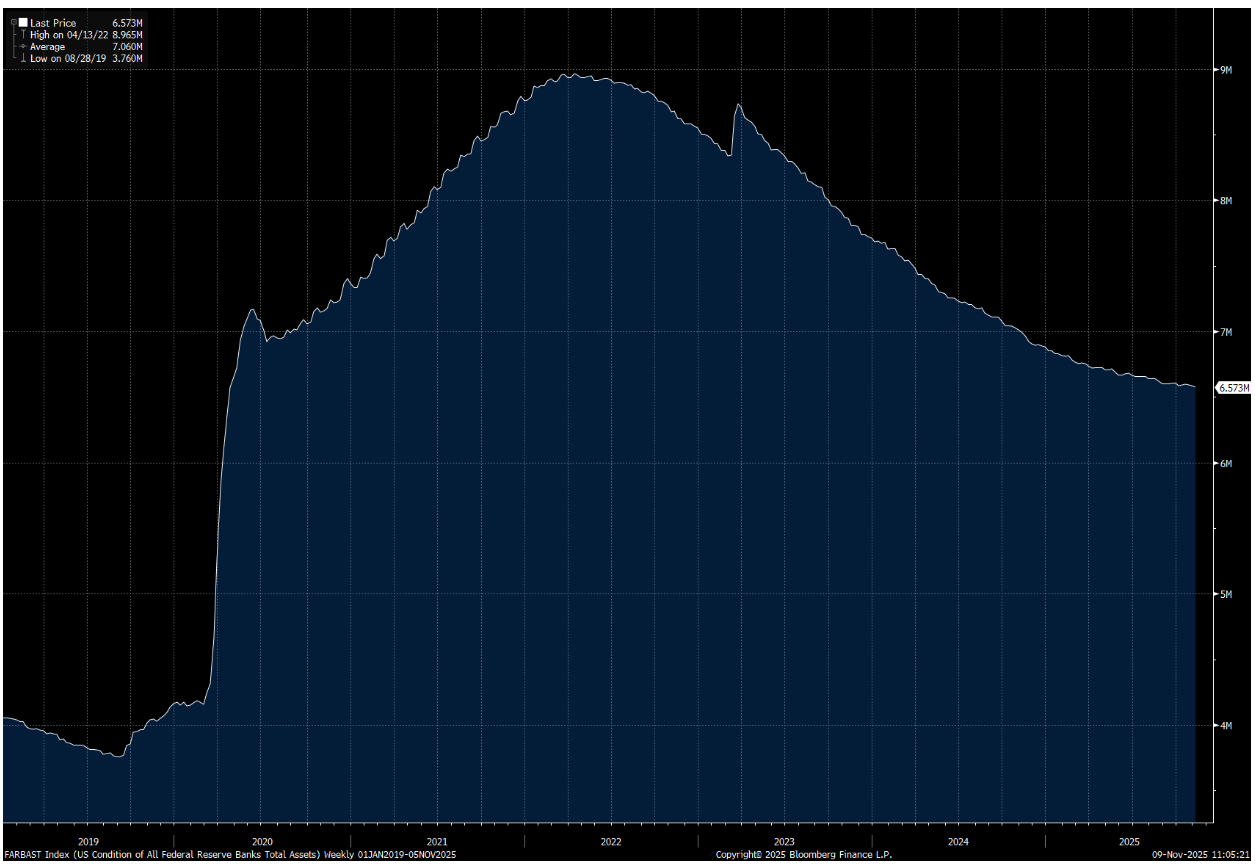

Figure 30-year Mortgage Rate 2019 – 2023 FRED

https://fred.stlouisfed.org/series/MORTGAGE30US#

Figure 5: Federal Reserve Holdings of MBS via Bloomberg 2019 – 2025. We can see how holdings start at around $1.4 tn before the start of QE in March 2020 and go to around $2.7 tn by March of 2022. They purchased roughly 1.33 tn of MBS during the period.

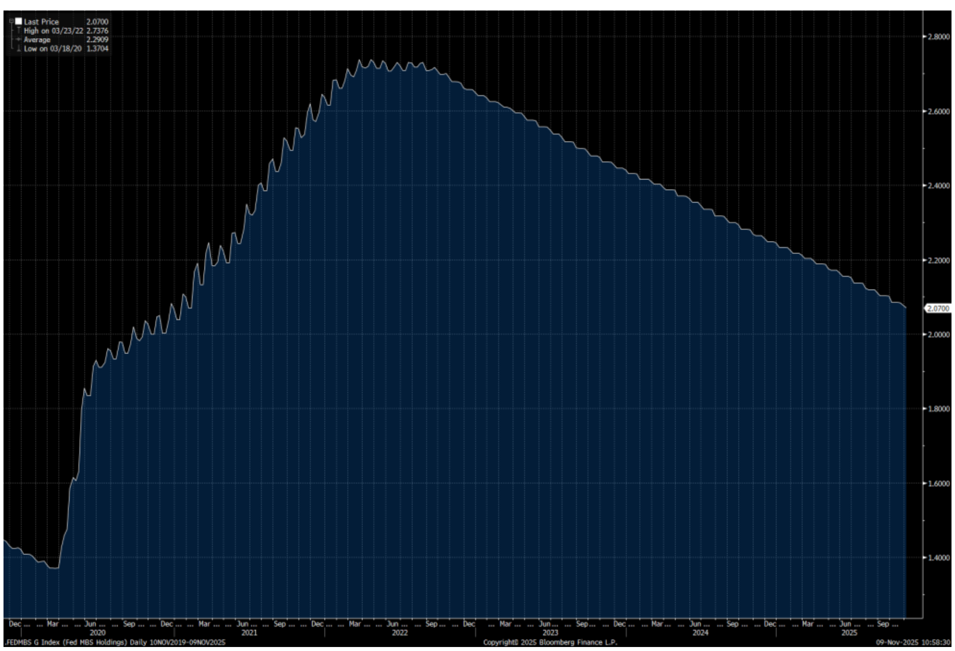

Figure 6: MBS vs 10-year Treasuries 2019 – 2025. Notice the sharp spike in March 2020 and the decline through mid-to-late 2021. The Fed ended QE in March 2022 and started Quantitative Tightening in June 2022.

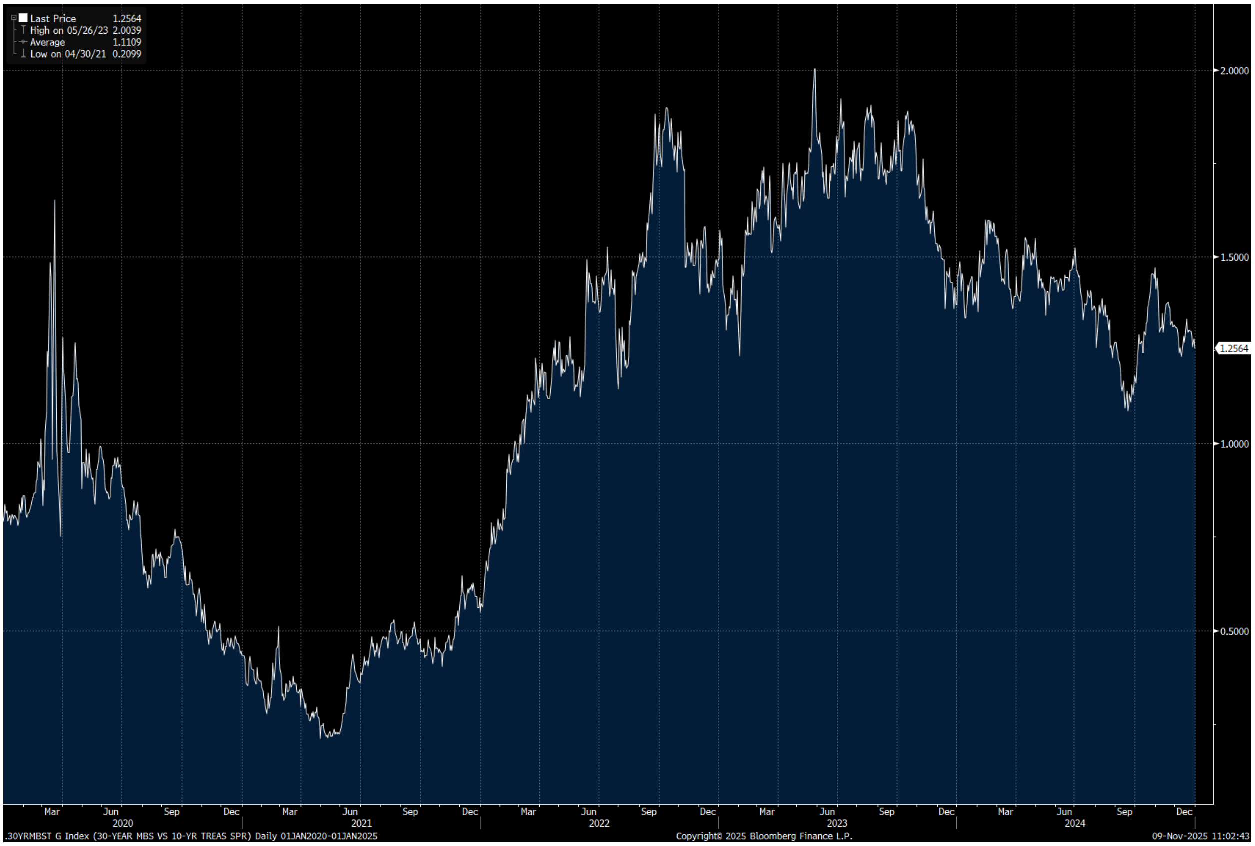

Figure 7: Fannie and Freddie Refinance Index data via Bloomberg. We can see during the 2020 – 2021 period the refinance index was almost double the next closest period. Data from 2005 – 2025.

Figure 8: FRED Data showing 30-year mortgage rates and 10-year treasuries. Notice the trough is during COVID (2020) for data from 1960 – 2025. Underlying data is from Freddie Mac for the 30-year mortgage rates.

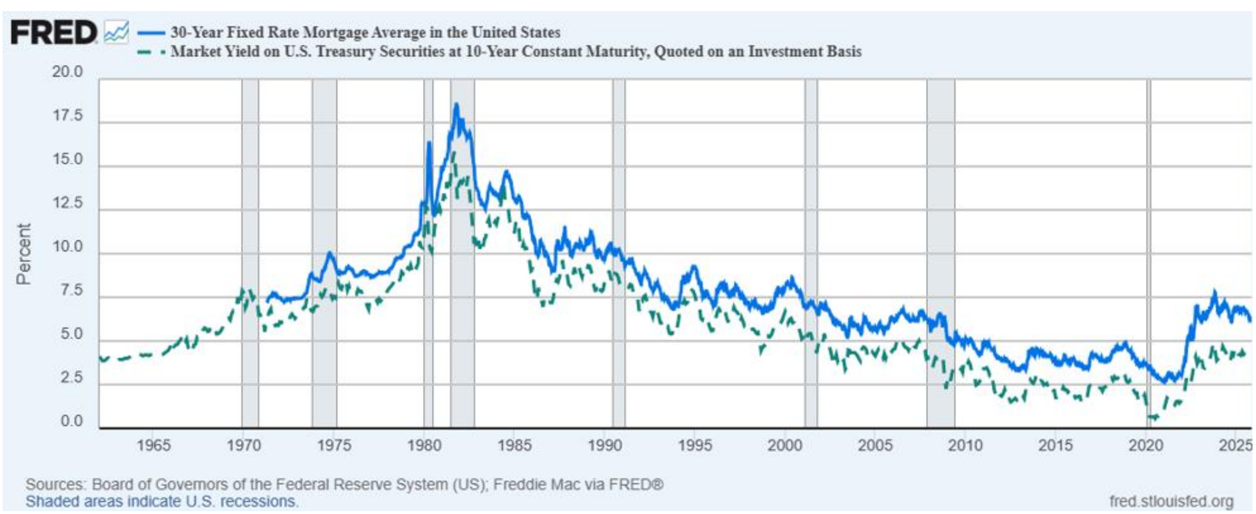

Figure 9: Home price data 2000 – 2025 via Zillow; note the rise above the trend line in 2020.

https://www.zillow.com/research/data/

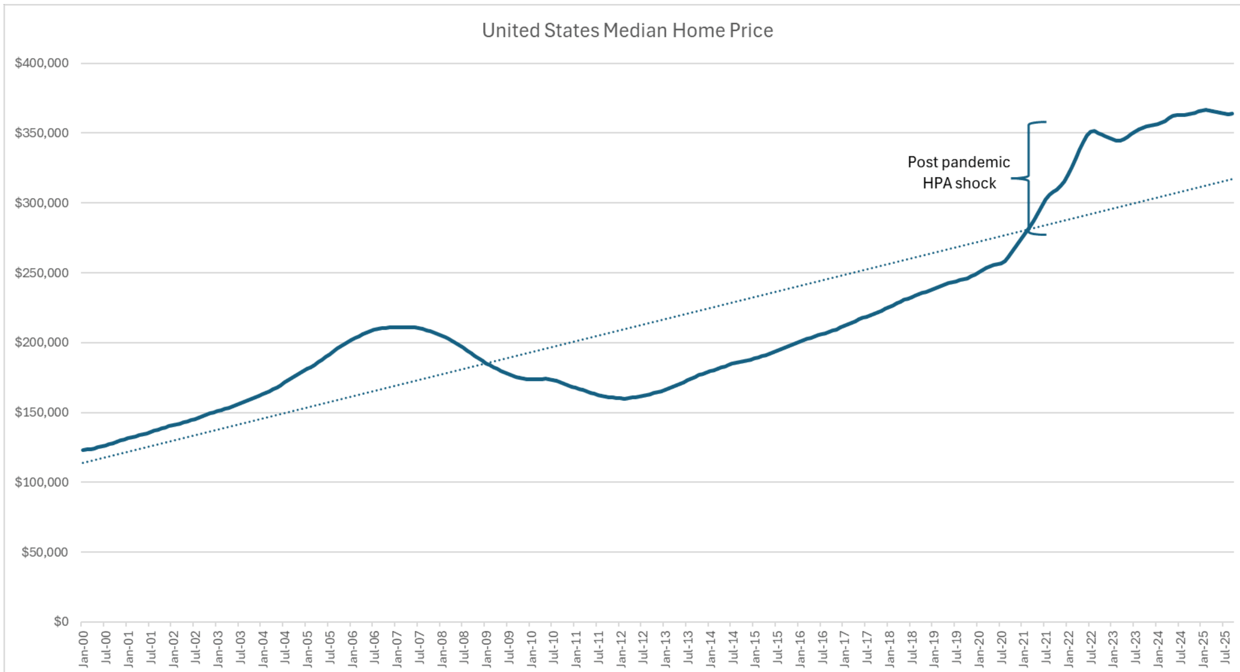

Figure 10: Quarterly home price data by 30-year fixed mortgage rate. Notice as mortgage rates increase on the x-axis, quarterly home prices trend down. The sharpest increases in quarterly home prices are from April 2020 – March 2022, when mortgage rates were below 3%. Data is a screenshot from this Brookings paper: https://www.brookings.edu/articles/quantitative-easing-and-housing-inflation-post-covid/.

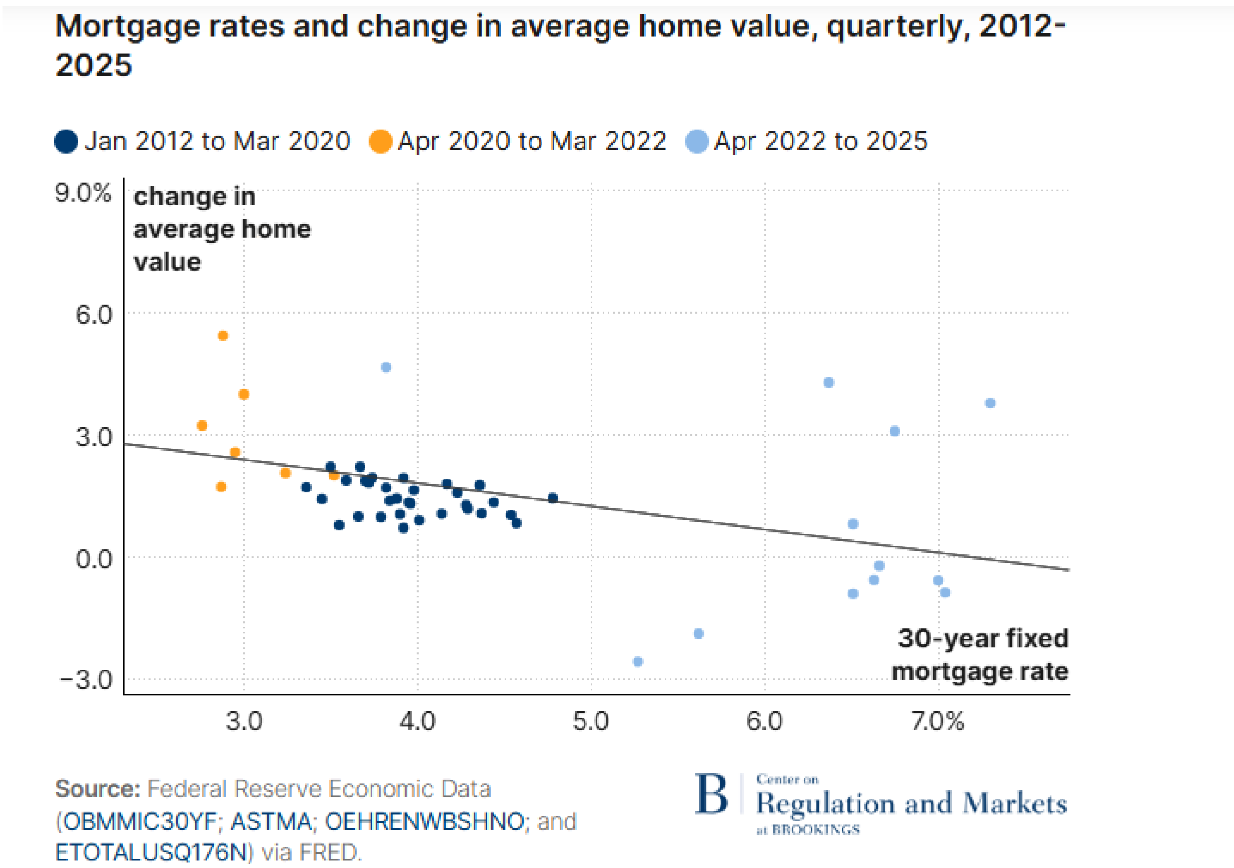

Figure 11: Existing Home Sales data via Bloomberg 1998 – 2025, from National Association of Realtors. Home Sales peaked in 2021, nearing record sales volumes to the 2005 – 2006 period

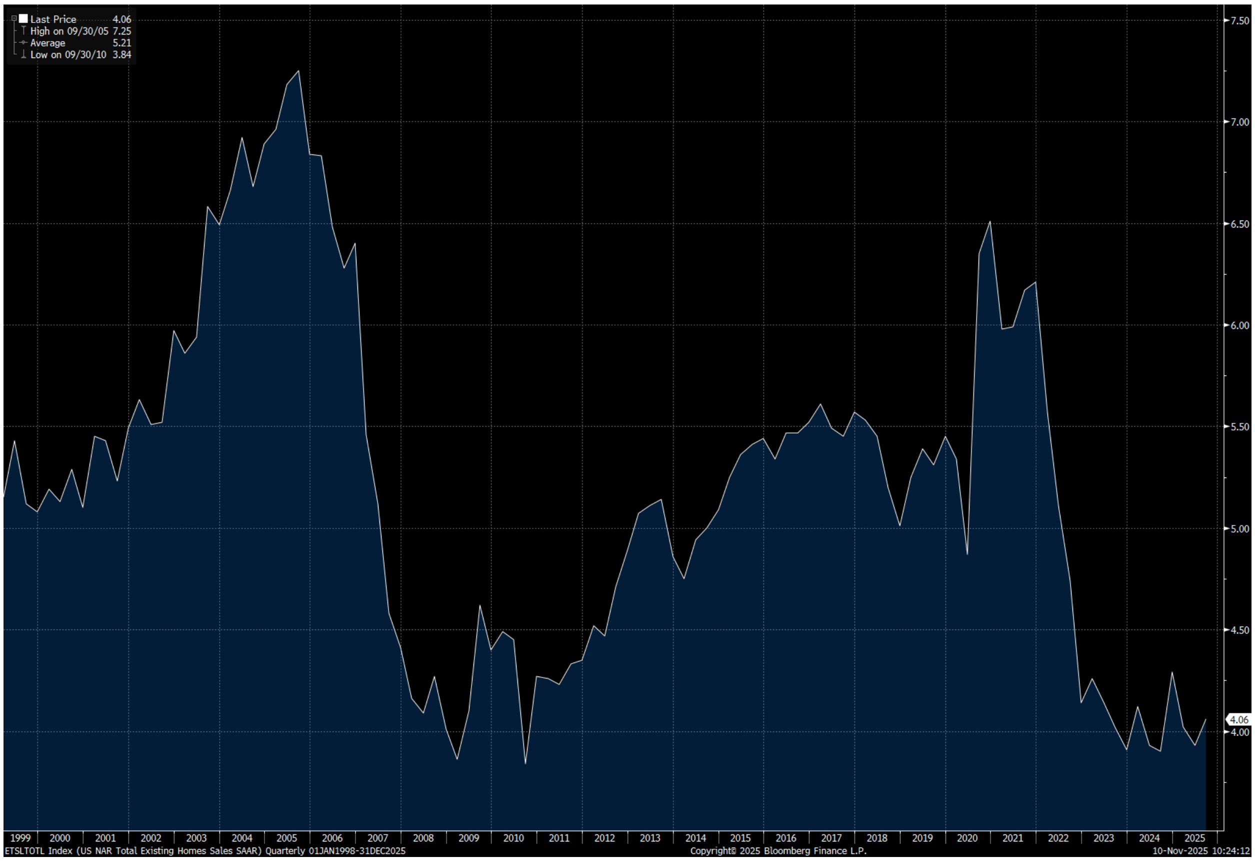

Figure 12: National Home Sales Seasonaly Adjusted – Data via Redfin. Home Sales fall close to 40% from May 2022 to November 2022, as quantitative tightening gets announced.

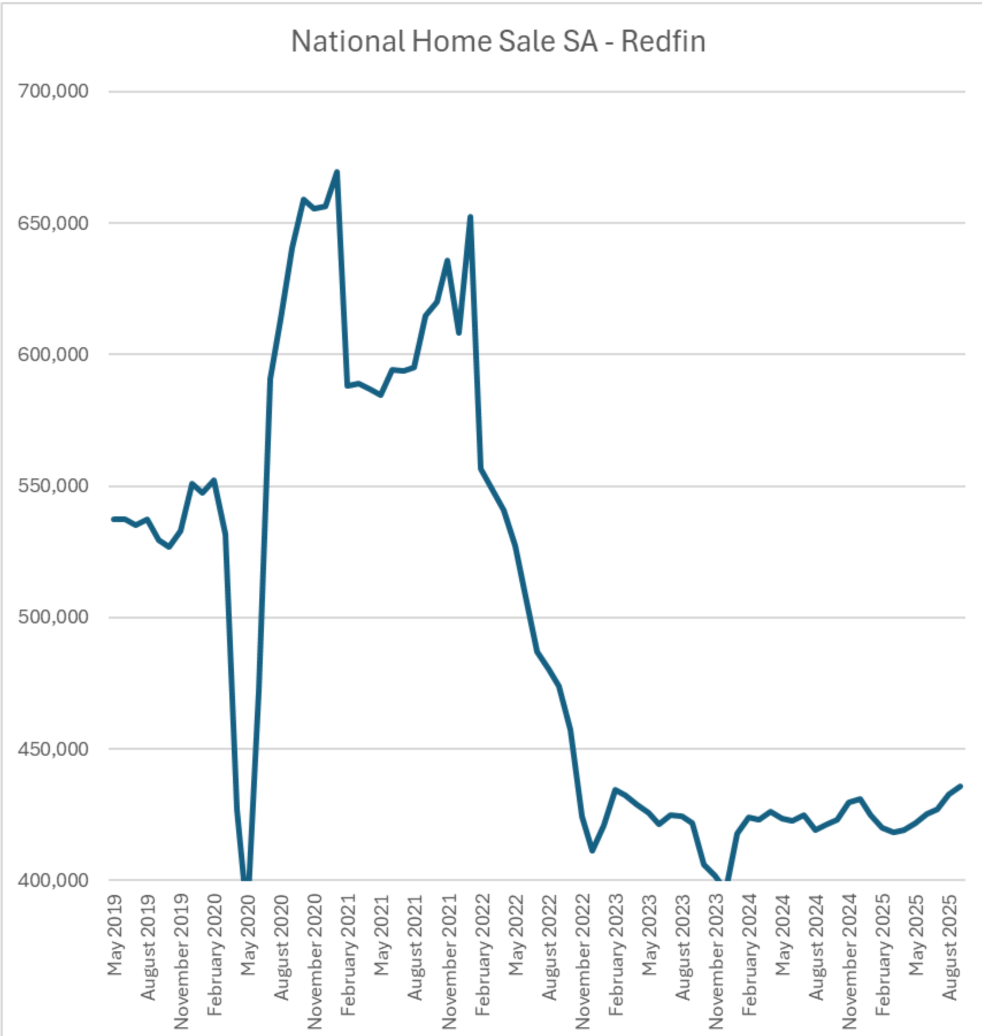

Figure 13: Aggregate Fed Balance Sheet Composition 2021 – March 2022; Data via Bloomberg.

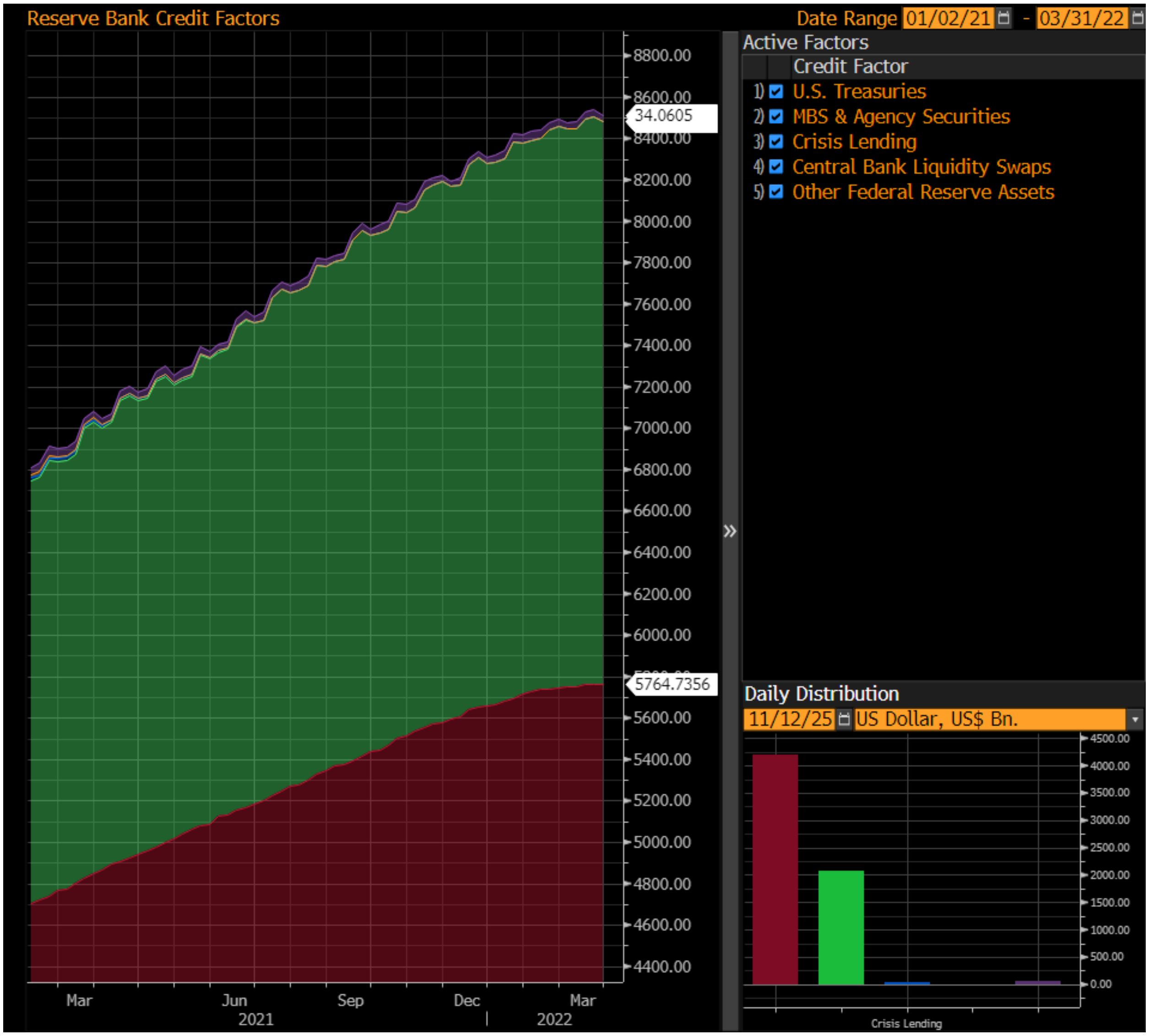

Figure 14: CPI year-over-year percentage change. Data via BLS, Bloomberg. 2019-2025. CPI Reached 4.2% in April 2021.

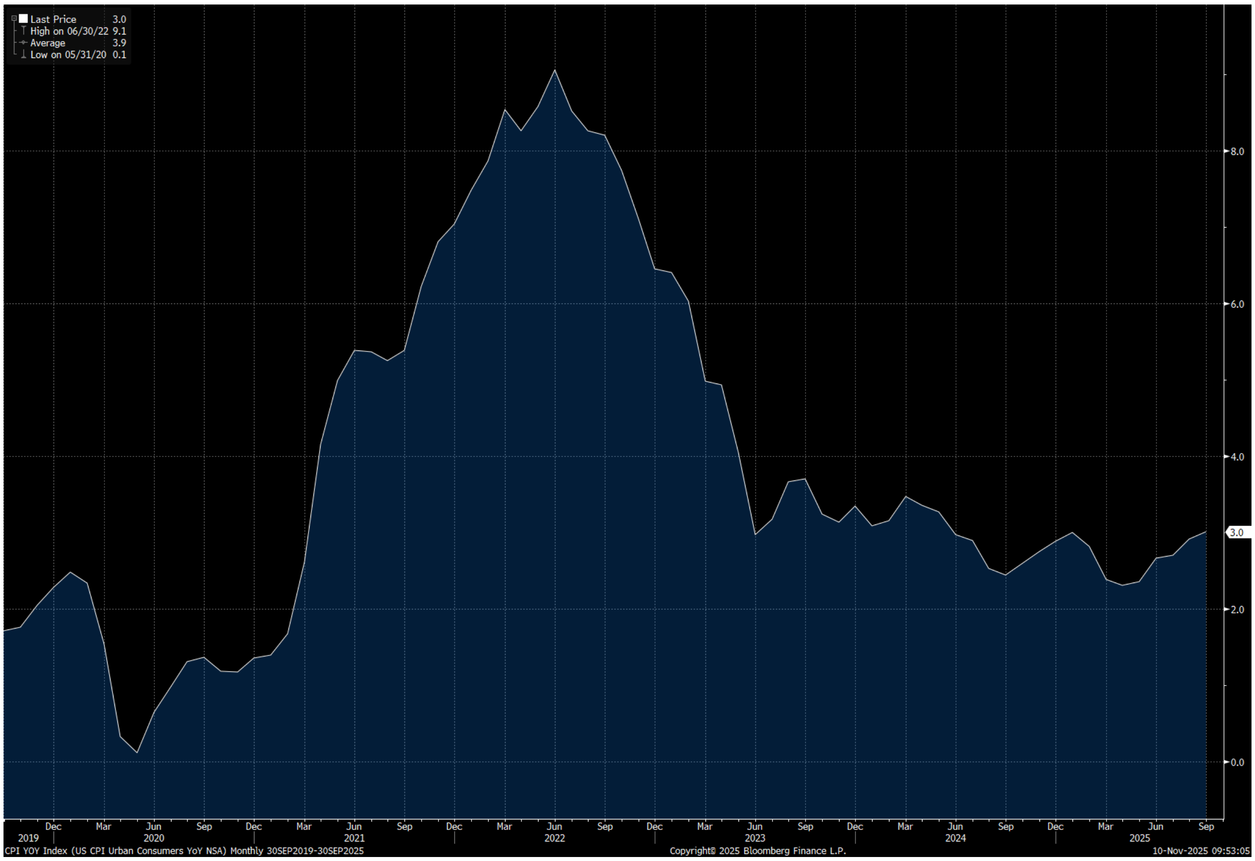

Figure 15: Shows the US Unemployment Rate (UE) from data via Bloomberg. The data is from 2019 – 2025. In 2021, as the economy reopened, we can see UE falling. By December of 2021, the UE was 4.1% signaling close to full employment.

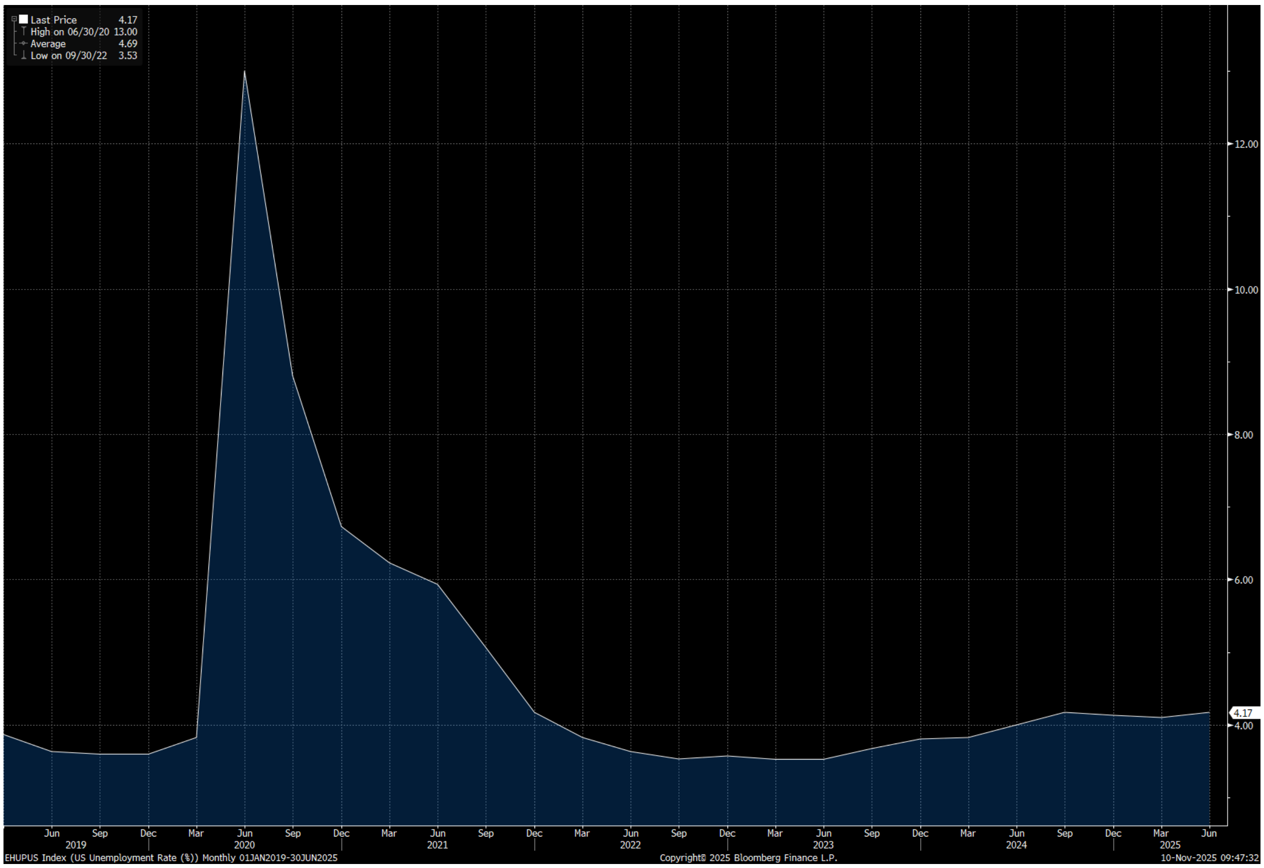

Figure 16: Shows the Jolt Job Openings from data via Bloomberg. The data is from 2019 – 2025. In 2021, we see labor demand pick-up considerably with opening going from 6.7mm to start 21' to 11mm to end 21'.

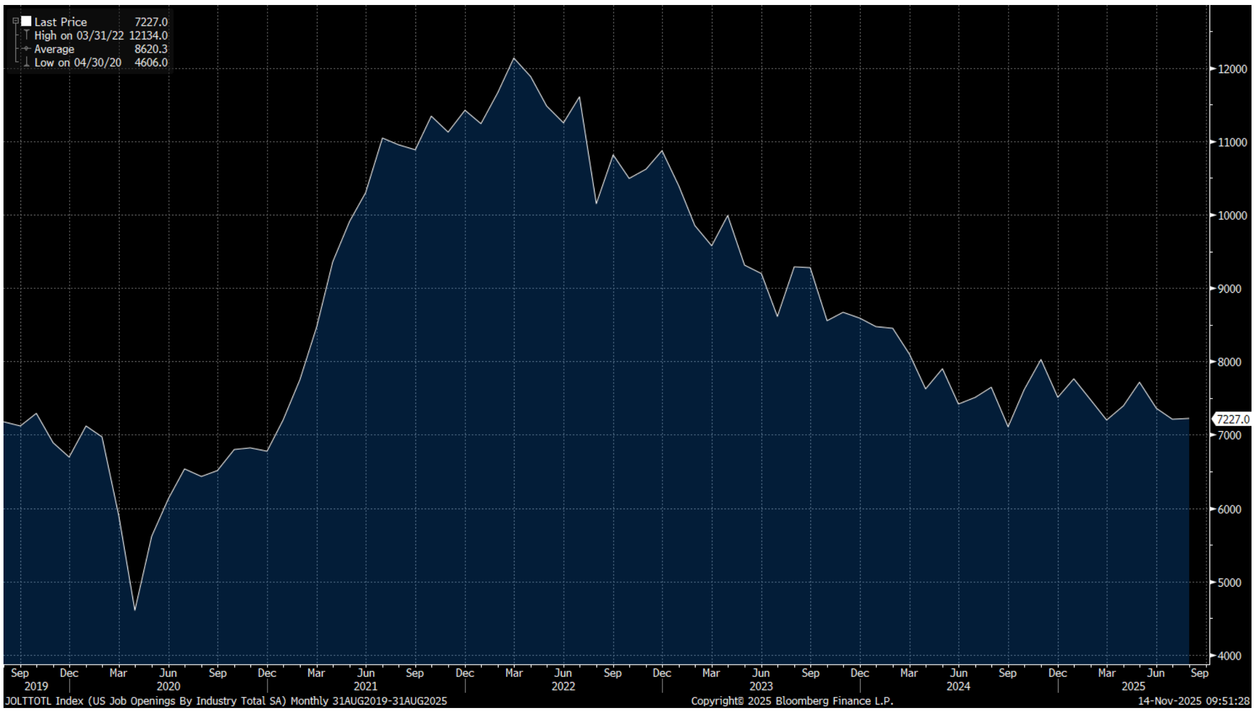

Figure 17: Home price year-over-year appreciate 2000 – 2025. Data via Zillow.

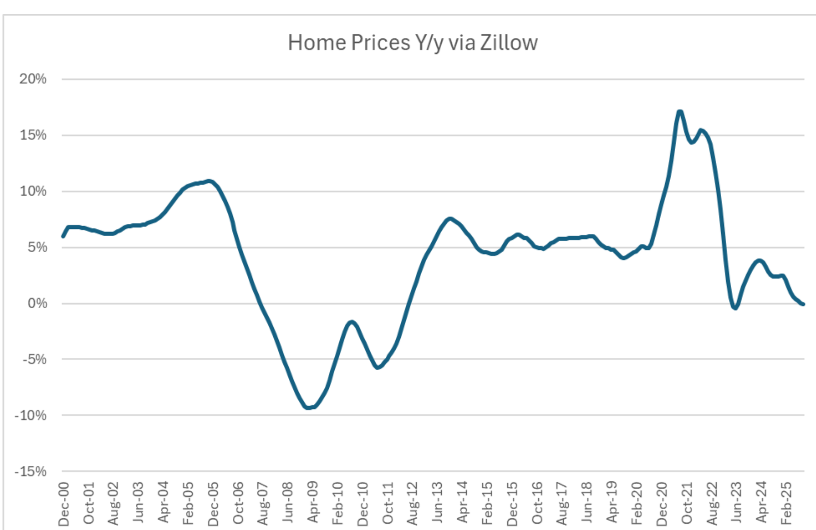

Figure 18: Affordability calculated as annual mortgage payment for an 80 LTV loan on the median home price in region divided by MSA median income. Data via Zillow and BEA, 2024 and 20225 Income estimated. No insurance or tax costs assumed.

Mortgage Payment Relative to Median Income

MSA	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
New York, NY	50%	55%	53%	49%	43%	38%	36%	29%	28%	31%	29%	29%	30%	31%	32%	28%	26%	31%	41%	44%	48%
Los Angeles, CA	74%	86%	86%	75%	56%	51%	46%	37%	39%	47%	45%	42%	46%	51%	53%	45%	42%	53%	69%	74%	79%
Chicago, IL	32%	36%	35%	33%	28%	25%	22%	17%	16%	18%	17%	17%	19%	19%	19%	17%	15%	18%	24%	26%	28%
Dallas, TX	23%	23%	23%	21%	20%	19%	17%	14%	14%	16%	16%	17%	19%	21%	21%	19%	17%	22%	31%	31%	32%
Houston, TX	21%	22%	21%	19%	18%	18%	16%	13%	12%	14%	15%	17%	18%	18%	18%	17%	15%	19%	26%	26%	27%
Washington, DC	42%	51%	50%	43%	34%	31%	29%	25%	26%	30%	28%	28%	30%	30%	30%	27%	25%	30%	39%	40%	43%
Philadelphia, PA	27%	32%	32%	30%	26%	24%	22%	18%	17%	18%	17%	17%	18%	19%	19%	17%	16%	19%	26%	28%	30%
Miami, FL	36%	45%	48%	41%	31%	24%	21%	17%	18%	22%	22%	24%	25%	26%	26%	23%	21%	26%	37%	40%	42%
Atlanta, GA	27%	29%	30%	29%	25%	21%	18%	14%	14%	17%	17%	17%	19%	20%	21%	19%	18%	24%	34%	35%	36%
Boston, MA	48%	49%	44%	40%	34%	32%	30%	25%	25%	29%	26%	27%	29%	30%	30%	26%	25%	29%	40%	43%	46%
Phoenix, AZ	34%	48%	47%	42%	30%	25%	21%	17%	20%	25%	23%	24%	26%	27%	28%	24%	23%	33%	42%	43%	43%
San Francisco, CA	61%	69%	66%	59%	49%	44%	40%	31%	33%	41%	40%	43%	45%	47%	48%	40%	35%	46%	56%	56%	59%
Riverside, CA	61%	75%	78%	65%	40%	34%	32%	27%	27%	37%	36%	36%	39%	44%	44%	37%	35%	48%	65%	67%	70%
Detroit, MI	24%	25%	24%	21%	16%	12%	11%	9%	9%	12%	12%	12%	14%	15%	16%	14%	13%	16%	21%	22%	24%
Seattle, WA	41%	47%	50%	48%	42%	37%	32%	24%	23%	27%	25%	27%	31%	33%	33%	29%	28%	36%	45%	46%	50%
Minneapolis, MN	32%	35%	34%	30%	26%	24%	21%	16%	17%	20%	18%	19%	21%	22%	22%	20%	18%	22%	29%	29%	30%
San Diego, CA	72%	79%	75%	64%	49%	46%	43%	35%	35%	43%	41%	43%	47%	50%	49%	41%	39%	52%	68%	74%	79%
Tampa, FL	28%	36%	38%	32%	24%	19%	17%	14%	14%	17%	17%	18%	20%	22%	22%	20%	18%	25%	36%	37%	37%
Denver, CO	36%	37%	35%	32%	31%	30%	27%	22%	22%	25%	25%	29%	31%	32%	32%	28%	25%	31%	41%	42%	43%
Baltimore, MD	33%	40%	41%	38%	32%	29%	26%	22%	21%	24%	22%	22%	23%	24%	23%	20%	19%	23%	29%	31%	32%
St. Louis, MO	24%	25%	25%	23%	21%	19%	18%	14%	14%	15%	14%	14%	15%	15%	15%	13%	12%	14%	19%	20%	21%
Orlando, FL	34%	46%	50%	43%	32%	24%	21%	17%	18%	22%	21%	23%	25%	26%	27%	24%	22%	29%	41%	43%	43%
Charlotte, NC	25%	25%	25%	23%	23%	22%	20%	15%	16%	18%	16%	17%	19%	20%	20%	18%	17%	23%	32%	33%	34%
San Antonio, TX	23%	24%	25%	24%	23%	21%	19%	16%	15%	17%	17%	18%	20%	20%	21%	19%	18%	22%	31%	30%	30%
Portland, OR	36%	44%	48%	45%	39%	35%	31%	24%	24%	29%	27%	30%	34%	34%	34%	30%	27%	35%	45%	45%	47%